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Studierichting: Informatica
Oefeningenreeks 4A nr. 42.6

$$\int \frac{x}{2x^2 - 3x + 1} dx$$

$$2x^2 - 3x + 1 = 0$$

$$\Delta = 9 - 8 = 1$$

$$x_1 = \frac{3+1}{4} = 1$$

$$x_2 = \frac{3-1}{4} = \frac{1}{2}$$

$$= \int \frac{x}{(x-1)(2x-1)} dx$$

$$\frac{x}{(x-1)(2x-1)} = \frac{A}{(x-1)} + \frac{B}{(2x-1)}$$

$$\Rightarrow A = \frac{x}{(2x-1)} = 1 \text{ met } x = 1$$

$$\Rightarrow B = \frac{x}{(x-1)} = -1 \text{ met } x = \frac{1}{2}$$

$$= \int \frac{x}{(x-1)(2x-1)} dx = \int \frac{1}{(x-1)} dx - \int \frac{1}{(2x-1)} dx$$

$$= \ln|x-1| - \frac{1}{2} \ln|2x-1| + c$$

References